Lab Assignment 2

Title: In a company, employee salaries are stored in a list as floating-point numbers. Write a Python program that sorts the employee salaries in ascending order using the following two algorithms:
• Selection Sort: Sort the salaries using the selection sort algorithm.
• Bubble Sort: Sort the salaries using the bubble sort algorithm.
After sorting the salaries, the program should display top five highest salaries in the company

Program code:

```
#include<stdio.h>
void bubbleSort(float arr[], int n){
   printf("\n========Bubble Sort========");

   int i, j;
   for(i=(n-1); i>=1; i--) {
      for(j=1; j<=i; j++) {
         if(arr[j-1]>arr[j]) {
            float temp = arr[j];
            arr[j] = arr[j-1];
            arr[j-1] = temp;
         }
      }
   }
   printf("\n\nThe sorted salaries are: ");
   for(i=0; i<n; i++){
      printf("%.2f ",arr[i]);
   }
}
void selectionSort(float arr[], int n){
   printf("\n========Selection Sort========");

   int i, j;
   for(i=(n-1); i>=1; i--) {
      int maxIdx=1;
      for(j=0; j<=i; j++) {
         if(arr[j] >= arr[maxIdx]){
            maxIdx = j;
         }
      }
      float temp = arr[i];
      arr[i] = arr[maxIdx];
      arr[maxIdx] = temp;
```

```
    }
    printf("\n\nThe sorted salaries are: ");
    for(i=0; i<n; i++){
        printf("%.2f ",arr[i]);
    }
}
int main(){
    int n, i, j;
    float arr[10];
    printf("\nEnter total no. of employees (max 10): ");
    scanf("%d",&n);

    printf("\nEnter employee salary in thousands: (Ex - 35.4)");
    for(i=0;i<n;i++)
    {
        printf("\nEnter salary of employee no. %d: ",i+1);
        scanf("%f",&arr[i]);
    }
    printf("\nEntered salaries are: ");
    for(i=0;i<n;i++)
    {
        printf("%.2f ",arr[i]);
    }

    printf("\n");

    char ch;
    do{
        int choice;
        printf("\nWhich type of sorting do you want to perform?");
        printf("\n1. Bubble Sort");
        printf("\n2. Selection Sort");
        printf("\nEnter choice (1 or 2): ");
        scanf("%d",&choice);

        switch(choice) {
            case 1:
                bubbleSort(arr, n);
                break;
            case 2:
                selectionSort(arr, n);
                break;
            default:
                printf("\nEnter valid choice");
```

```
        }

        printf("\nDo you want to continue?(y/n): ");
        scanf("  %c", &ch);
    }while(ch == 'y'|| ch == 'Y');
    return 0;
}
```

Output:

```
Enter total no. of employees (max 10): 5

Enter employee salary in thousands: (Ex - 35.4)
Enter salary of employee no. 1: 35.4
Enter salary of employee no. 2: 28.5
Enter salary of employee no. 3: 42.6
Enter salary of employee no. 4: 31.2
Enter salary of employee no. 5: 25.8

Entered salaries are: 35.40 28.50 42.60 31.20 25.80

Which type of sorting do you want to perform?
1. Bubble Sort
2. Selection Sort
Enter choice (1 or 2): 1

========Bubble Sort========

The sorted salaries are: 25.80 28.50 31.20 35.40 42.60
Do you want to continue?(y/n): y

Which type of sorting do you want to perform?
1. Bubble Sort
2. Selection Sort
Enter choice (1 or 2): 2

========Selection Sort========

The sorted salaries are: 25.80 28.50 31.20 35.40 42.60
Do you want to continue?(y/n): n
```